

# Unit-1: Logics

## 1. Definition and Key Terms:

**1 Logic** : Logic is the branch of mathematics that studies the principles of valid reasoning and methods to determine whether a statement is true or false.

 **Understanding :**

**Logic helps us decide:**

- Is the reasoning correct?
- Does the conclusion follow from given facts?

**2 Statement** : A statement is a declarative sentence that has a definite truth value, either true or false, but not both simultaneously.

 **Understanding:**

**It must:**

- Be a complete sentence
- Have fixed truth value

✓ **Example:** " $2 + 2 = 4$ "

✗ **Not a statement:** "How are you?"

**3 Truth Values:** Truth values are the possible values assigned to a statement, namely True (T) or False (F).

 **Understanding:**

**Every statement in logic can only be:**

- True (T)
- False (F)

There is no middle value in classical logic.

**4 Proposition** : A proposition is a statement that has a definite truth value, either true or false.

 **Understanding:**

Every proposition is a statement.

Every statement in propositional logic is called a proposition.

Example:

" $5 > 3$ "  $\rightarrow$  Proposition (True)

**5 Propositional Logic** : Propositional logic is the study of propositions and logical connectives used to form compound statements whose truth values depend on the, truth values of their components.

 **Understanding:**

It deals with:

- Whole statements
- Combining them using AND, OR, NOT, etc.

**6 Tautology** : A tautology is a compound statement that is true for all possible truth values of its components.

 **Understanding:**

Example:  $p \vee \neg p$

Where "**p**" is propositional variable.

No matter what p is  $\rightarrow$  always True.

Used to prove logical laws.

**7 Notation :** Notation in logic refers to the system of symbols used to represent logical statements, operations, and relationships in a concise and precise manner.

#### **Understanding**

Instead of writing full sentences, we use symbols.

Example:

- $p$  = "It is raining"
- $q$  = "I carry umbrella"

Symbols help in:

- Writing formulas
- Making truth tables
- Proving equivalence

**8 Connectives :** Logical connectives are symbols used to combine one or more propositions to form a compound statement.

#### **Understanding:** They connect statements together.

Example:

If

$p$  = It is raining

$q$  = It is cold

Then:

- $p \wedge q$
- $p \vee q$
- $\neg p$
- $p \rightarrow q$

#### **Main Logical Connectives**

| Symbol                | Name          | Meaning                |
|-----------------------|---------------|------------------------|
| $\neg p$ or $\sim p$  | Negation      | Not $p$                |
| $p \wedge q$          | Conjunction   | $p$ AND $q$            |
| $p \vee q$            | Disjunction   | $p$ OR $q$             |
| $p \rightarrow q$     | Implication   | If $p$ then $q$        |
| $p \leftrightarrow q$ | Biconditional | $p$ if and only if $q$ |

**9 Truth Table :** A truth table is a tabular representation that lists all possible combinations of truth values of propositions and shows the resulting truth value of a compound statement.

#### **Understanding**

Truth table helps to:

- Check validity
- Identify tautology
- Test equivalence

#### **Example Truth Table**

For  $p \wedge q$ :

| $p$ | $q$ | $p \wedge q$ |
|-----|-----|--------------|
| T   | T   | T            |
| T   | F   | F            |
| F   | T   | F            |
| F   | F   | F            |

Explanation: AND is true only when both are true. **10 Predicate:** A predicate is a statement containing one or more variables that becomes a proposition when specific values are assigned to the variables.

### Understanding

Example:

$P(x): x > 5$

If  $x = 7 \rightarrow \text{True}$

If  $x = 3 \rightarrow \text{False}$

So predicate becomes proposition when variable gets value. **11 Predicate Logic** : Predicate logic is an extension of propositional logic that uses predicates, variables, and quantifiers to express statements about objects and their properties.

### Understanding

It deals with:

- Variables
- Objects
- Quantities (All, Some)

Example:

"All students passed."

**12 Statement Function:** A statement function is an expression containing one or more variables that becomes a proposition when specific values are assigned to the variables.

### Understanding

It does not have a truth value until a value is given.

Example:

$P(x): x > 5$

If  $x = 7 \rightarrow \text{True}$

If  $x = 3 \rightarrow \text{False}$

**1 Variable** : A variable is a symbol used to represent an arbitrary element from a specified domain or universe of discourse.

### Understanding

Variables usually represented by:

$x, y, z$

They stand for objects in a domain.

Example:

$x \in \text{Natural Numbers}$

**2 Quantifiers** : Quantifiers are symbols used in predicate logic to indicate the number of elements in the domain for which a statement is true.

#### Types of Quantifiers

1. **Universal Quantifier ( $\forall$ )**  
Means "for all"
2. **Existential Quantifier ( $\exists$ )**  
Means "there exists"

#### Example

$\forall x P(x)$

"For all  $x$ ,  $P(x)$  is true."

$\exists x P(x)$

"There exists at least one  $x$  such that  $P(x)$  is true."

**3 Free and Bound Variables** : A free variable is a variable that is not quantified in a statement, whereas a bound variable is a variable that is quantified by a quantifier.

### Understanding

Example:

$P(x) = x > 5$

Here x is **free**.

Now consider:

$\forall x (x > 5)$

Here x is **bound** because it is controlled by  $\forall$ .

**4 Universe of Discourse (Domain of Discourse):** The universe of discourse is the set of all elements over which the variables of a predicate are defined.

#### **Understanding**

It defines the “range” of possible values.

Example:

If  $U = \{1, 2, 3, 4, 5\}$

Then x can only take values from this set.

Without defining universe, predicate logic is incomplete.

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 **Tautology** : A tautology is a compound logical statement that is true for all possible combinations of truth values of its component propositions.

**Key Point:** It is **always true**, no matter what the truth values are.

**Example:**  $p \vee \neg p$

This is always True.

 **Contradiction:** A contradiction is a compound logical statement that is false for all possible combinations of truth values of its component propositions.

**Key Point:** It is **always false**, no matter what the truth values are.

**Example:**  $p \wedge \neg p$

This is always False.

## **2. Propositional Logic:**

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Now we move properly into **Propositional Logic** as per GTU syllabus.

We will cover:

1. Statement formulas
2. Logical connectives (deep meaning)
3. Truth tables (construction method)
4. Well-formed formulas
5. Tautology / Contradiction / Contingency
6. Logical equivalence
7. Duality law
8. Tautological implication

**1 Statement Formulas :** A statement formula is a compound logical expression formed by combining propositions using logical connectives.

#### **Example**

Let:

$p$  = It is raining

$q$  = I carry umbrella

Statement formula:

$(p \wedge q) \rightarrow p$

It is built using:

- Variables ( $p, q$ )
- Connectives ( $\wedge, \rightarrow$ )

## **2 Logical Connectives – Deep Understanding**

### **(A) Negation ( $\neg p$ or $\sim p$ )**

Meaning: "Not p"

Truth Table:

| p | $\neg p / \sim p$ |
|---|-------------------|
| T | F                 |
| F | T                 |

It simply reverses truth value.

**(B) Conjunction ( $p \wedge q$ )**

Meaning: "p AND q"

True only when both are true.

| p | q | $p \wedge q$ |
|---|---|--------------|
| T | T | T            |
| T | F | F            |
| F | T | F            |
| F | F | F            |

**(C) Disjunction ( $p \vee q$ )**

Meaning: "p OR q"

True when at least one is true.

| p | q | $p \vee q$ |
|---|---|------------|
| T | T | T          |
| T | F | T          |
| F | T | T          |
| F | F | F          |

**(D) Implication ( $p \rightarrow q$ ) (conditional proposition) ★ Important**

Meaning: "If p then q"

Truth Table:

| p | q | $p \rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

Important Rule:

$p \rightarrow q$  is false only when:

p is True and q is False.

Otherwise it is True.

Reason: If condition does not happen, rule is not broken.

**(E) Biconditional ( $p \leftrightarrow q$ )**

Meaning: "p if and only if q"

True when both have same truth value.

| p | q | $p \leftrightarrow q$ |
|---|---|-----------------------|
|---|---|-----------------------|

|   |   |   |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | T |

### 3 How to Construct Truth Table (Step-by-Step Method)

If n propositions:

**Number of rows =  $2^n$**

Example:

For p and q:

$2^2 = 4$  rows

For p, q, r:

$2^3 = 8$  rows

Procedure:

1. Write combinations of T and F.
2. Solve inside brackets first.
3. Move step by step.

Example : Construct truth table for:  $(p \wedge q) \rightarrow p$

#### ✓ Step 1: Count Variables

Variables: p, q

Number of rows =  $2^n$

Here n = 2

$2^2 = 4$  rows

#### ✓ Step 2: Write All Combinations of p and q

| p | q |
|---|---|
| T | T |
| T | F |
| F | T |
| F | F |

#### ✓ Step 3: Solve Inside Bracket First ( $p \wedge q$ )

AND rule: True only if both are true.

| p | q | $p \wedge q$ |
|---|---|--------------|
| T | T | T            |
| T | F | F            |
| F | T | F            |
| F | F | F            |

#### ✓ Step 4: Apply Implication Rule

Implication rule:

$p \rightarrow q$  is false only when:

p = T and q = F

Now apply to:

$(p \wedge q) \rightarrow p$

| p | q | $p \wedge q$ | $(p \wedge q) \rightarrow p$ |
|---|---|--------------|------------------------------|
|---|---|--------------|------------------------------|

|   |   |   |   |
|---|---|---|---|
| T | T | T | T |
| T | F | F | T |
| F | T | F | T |
| F | F | F | T |

#### ✓ Step 5: Final Observation

Last column = All True

Therefore:

$(p \wedge q) \rightarrow p$  is a **Tautology**

#### 4 Well-Formed Formula (WFF)

**Definition :** A well-formed formula is a logical expression that is constructed according to the rules of syntax of propositional logic.

Rules:

- Propositions are WFF.
- If  $p$  is WFF, then  $\neg p$  is WFF.
- If  $p$  and  $q$  are WFF, then  $(p \wedge q)$ ,  $(p \vee q)$ ,  $(p \rightarrow q)$  are WFF.

Incorrect example:

$p \wedge \rightarrow q$  ✗

#### 5 Types of Compound Statements

**(A) Tautology :** Always True.

Example:  $p \vee \neg p$

**(B) Contradiction :** Always False.

Example:  $p \wedge \neg p$

**(C) Contingency**

Sometimes true, sometimes false.

Example:  $p \wedge q$

**6 Logical Equivalence:** Two statements are logically equivalent if they have identical truth values for all possible combinations.

Notation:  $p \equiv q$

Example:  $p \rightarrow q \equiv \neg p \vee q$

This is very important identity.

#### 7 Laws of Propositional Logic

| No. | Law Name                                  | Formula                                                     |
|-----|-------------------------------------------|-------------------------------------------------------------|
| 1   | Idempotent Law ( $\vee$ )                 | $p \vee p \equiv p$                                         |
| 2   | Idempotent Law ( $\wedge$ )               | $p \wedge p \equiv p$                                       |
| 3   | Commutative Law ( $\vee$ )                | $p \vee q \equiv q \vee p$                                  |
| 4   | Commutative Law ( $\wedge$ )              | $p \wedge q \equiv q \wedge p$                              |
| 5   | Associative Law ( $\vee$ )                | $p \vee (q \vee r) \equiv (p \vee q) \vee r$                |
| 6   | Associative Law ( $\wedge$ )              | $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$        |
| 7   | Distributive Law ( $\wedge$ over $\vee$ ) | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ |
| 8   | Distributive Law ( $\vee$ over $\wedge$ ) | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$   |
| 9   | Double Negation Law                       | $\neg(\neg p) \equiv p$                                     |

|    |                   |                                              |
|----|-------------------|----------------------------------------------|
| 10 | De Morgan's Law   | $\neg(p \vee q) \equiv \neg p \wedge \neg q$ |
| 11 | De Morgan's Law   | $\neg(p \wedge q) \equiv \neg p \vee \neg q$ |
| 12 | Tautology Law     | $p \vee \neg p \equiv T$                     |
| 13 | Contradiction Law | $p \wedge \neg p \equiv F$                   |
| 14 | Absorption Law    | $p \vee (p \wedge q) \equiv p$               |
| 15 | Absorption Law    | $p \wedge (p \vee q) \equiv p$               |

### Derived Logical Formulas

| No. | Law / Formula Name                | Derived Formula                                                         |
|-----|-----------------------------------|-------------------------------------------------------------------------|
| 1   | Law of Excluded Middle            | $p \vee \neg p \equiv T$                                                |
| 2   | Law of Contradiction              | $p \wedge \neg p \equiv F$                                              |
| 3   | Identity Law ( $\vee$ with T)     | $p \vee T \equiv T$                                                     |
| 4   | Identity Law ( $\vee$ with F)     | $p \vee F \equiv p$                                                     |
| 5   | Identity Law ( $\wedge$ with T)   | $p \wedge T \equiv p$                                                   |
| 6   | Domination Law ( $\wedge$ with F) | $p \wedge F \equiv F$                                                   |
| 7   | Implication Law                   | $p \rightarrow q \equiv \neg p \vee q$                                  |
| 8   | Biconditional Law                 | $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ |

### 8 Duality Law:

**Definition:** The dual of a logical expression is obtained by interchanging  $\wedge$  with  $\vee$  and T with F.

Example:

Original:  $p \wedge T \equiv p$

Dual:  $p \vee F \equiv p$

### 9 Tautological Implication

**Definition:** A statement P tautologically implies Q if  $P \rightarrow Q$  is a tautology.

Notation:  $P \models Q$

Meaning: Whenever P is true, Q must be true.

Used to prove arguments.

### ◆ DNF and CNF

## DISCRETE MATHEMATICS

### # LAW OF DUALITY:-

**Definition:-** Two Statements A and A\* are said to be duals of each other if either one can be obtained from the other by replacing  $\wedge$  by  $\vee$ ,  $\vee$  by  $\wedge$ , T by F & F by T

- ①  $\vee$  is replaced by  $\wedge$
- ②  $\wedge$  is replaced by  $\vee$
- ③ T is replaced by F
- ④ F is replaced by T



$\wedge$  Conjunction  
 $\vee$  Disjunction



These are **standard forms** of logical expressions.

They are heavily used in:

- Digital circuits
- Boolean algebra
- AI rule systems
- Logic simplification

**1 Conjunctive Normal Form (CNF)** : A logical formula is said to be in Conjunctive Normal Form (CNF) if it is a conjunction (AND) of one or more disjunctions (ORs) of literals.

#### Simple Meaning

CNF = AND of ORs

Structure:

$$(A \vee B) \wedge (C \vee D)$$

Each bracket:

- Contains OR ( $\vee$ )
- Whole expression connected by AND ( $\wedge$ )

#### Example of CNF

$$(p \vee q) \wedge (\neg p \vee r)$$

✓ Each bracket is OR

✓ Brackets connected by AND

So it is CNF.

**2 Disjunctive Normal Form (DNF)** : A logical formula is said to be in Disjunctive Normal Form (DNF) if it is a disjunction (OR) of one or more conjunctions (ANDs) of literals.

#### Simple Meaning

DNF = OR of ANDs

Structure:  $(A \wedge B) \vee (C \wedge D)$

Each bracket:

- Contains AND ( $\wedge$ )
- Brackets connected by OR ( $\vee$ )

#### Example of DNF

$$(p \wedge q) \vee (\neg p \wedge r)$$

✓ Each bracket is AND

✓ Brackets connected by OR

So it is DNF.

#### What is a Literal?

A literal is:

- A proposition ( $p$ )
- OR its negation ( $\neg p$ )

Examples:

$p, \neg p, q, \neg r$

#### Difference Between CNF and DNF

| Basis          | CNF                            | DNF                              |
|----------------|--------------------------------|----------------------------------|
| Full Form      | Conjunctive Normal Form        | Disjunctive Normal Form          |
| Structure      | AND of ORs                     | OR of ANDs                       |
| Outer Operator | AND ( $\wedge$ )               | OR ( $\vee$ )                    |
| Inner Operator | OR ( $\vee$ )                  | AND ( $\wedge$ )                 |
| Example        | $(p \vee q) \wedge (r \vee s)$ | $(p \wedge q) \vee (r \wedge s)$ |

### 3 How to Convert to DNF (Using Truth Table)

Example: Construct DNF of:

$p \rightarrow q$

Step 1: Build truth table

| p | q | $p \rightarrow q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

Step 2: Take rows where result is TRUE.

Row 1:  $p=T, q=T \rightarrow (p \wedge q)$

Row 3:  $p=F, q=T \rightarrow (\neg p \wedge q)$

Row 4:  $p=F, q=F \rightarrow (\neg p \wedge \neg q)$

Step 3: Join with OR

DNF:  $(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$

#### 4 How to Convert to CNF (Using False Rows)

Take rows where result is FALSE.

From table:

Only row 2 is false ( $p=T, q=F$ )

CNF:  $(\neg p \vee q)$

Which simplifies to well-known identity:

$p \rightarrow q \equiv \neg p \vee q$

#### 5 Why CNF and DNF Matter?

In computer science:

- Digital circuit design uses DNF (Sum of Products)
- SAT solvers use CNF
- AI logic resolution works on CNF
- Boolean simplification uses DNF



#### Easy Memory Trick

CNF  $\rightarrow$  Conjunction first

DNF  $\rightarrow$  Disjunction first

DNF: OR of ANDs

CNF: AND of Ors

### 3. Predicate Logic

#### 1 What is Predicate Logic?

**Definition :** Predicate logic is an extension of propositional logic that uses predicates, variables, and quantifiers to express statements about objects and their properties.



#### Why Predicate Logic is Needed?

Propositional logic cannot express:

- All students passed.
- Some numbers are prime.
- Every person has a mother.

Because these involve:

- Objects
- Variables
- Quantity (All / Some)

Predicate logic solves this.

## 2 Predicate

**Definition :** A predicate is a statement containing one or more variables that becomes a proposition when specific values are assigned.

Example:

$$P(x): x > 5$$

$$\text{If } x = 7 \rightarrow \text{True}$$

$$\text{If } x = 2 \rightarrow \text{False}$$

**3 Statement Function:** A statement function is another name for predicate.

Example:

$$P(x) = x \text{ is even}$$

It becomes proposition when:

$$x = 4 \rightarrow \text{True}$$

**5 Quantifiers:** Quantifiers tell how many elements satisfy the predicate.

### (A) Universal Quantifier ( $\forall$ )

Symbol:  $\forall$

Meaning: "For all"

**Example:**

$$\forall x P(x)$$

"For all x, P(x) is true."

Example:

$$\forall x (x + 0 = x)$$

### (B) Existential Quantifier ( $\exists$ )

Symbol:  $\exists$

Meaning: "There exists"

**Example:**

$$\exists x P(x)$$

"There exists at least one x such that P(x) is true."

Example:

$$\exists x (x^2 = 4)$$

## 5 Universe of Discourse

**Definition :** The universe of discourse is the set of all elements over which variables are defined.

Example:

$$\text{If } U = \{1, 2, 3, 4, 5\}$$

Then x can only take values from this set.

Without universe, predicate logic is incomplete.

## 6 Free and Bound Variables

**Bound Variable :** A variable controlled by quantifier.

Example:

$\forall x (x > 0)$

x is bound.

**Free Variable :** A variable not controlled by quantifier.

Example:

$x > 0$

x is free.

Free variable  $\rightarrow$  Not complete proposition.

## 7 Negation of Quantifiers (Very Important)

**Rule 1:**

$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$

Meaning:

Not all  $\rightarrow$  There exists at least one not.

**Rule 2:**

$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$

Meaning: Not exists  $\rightarrow$  For all not.

## 8 Example Translation

English  $\rightarrow$  Logic

"All students passed."

Let:

$S(x)$ : x is student

$P(x)$ : x passed

Logical form:

$\forall x (S(x) \rightarrow P(x))$

Ex:- "There exists a student who failed."

Logical form:

$\exists x (S(x) \wedge \neg P(x))$

## ✚ Why Predicate Logic is Powerful?

It is used in:

- Database queries
- Artificial Intelligence
- Knowledge representation
- Automated reasoning
- Formal verification

## ★ Video Based Learning

**Logic Playlist:** <https://youtube.com/playlist?list=PLIHmQ48sVguc9X-jdLC764jml0bCJq50&si=z9kXS17xVpNLkbFa/>

## Exercise Set:

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### ◆ PART 1: 12 Truth Table Practice Problems

Construct truth table and identify (Tautology / Contradiction / Contingency)

1.  $\neg(p \wedge q)$
2.  $p \rightarrow (q \vee p)$
3.  $(p \rightarrow q) \wedge p$
4.  $(p \vee q) \rightarrow p$
5.  $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
6.  $(p \wedge q) \rightarrow (p \vee q)$
7.  $(p \vee \neg p) \wedge q$
8.  $(p \rightarrow q) \vee (q \rightarrow p)$
9.  $\neg(p \rightarrow q)$
10.  $(p \wedge (p \rightarrow q)) \rightarrow q$
11.  $(p \vee q) \wedge (\neg p \wedge \neg q)$
12.  $(p \rightarrow (q \rightarrow r))$

Goal:

- Practice 2-variable and 3-variable cases.
- Identify final column nature.

### ◆ PART 2: 10 Logical Equivalence Simplifications

Use logical laws to simplify.

1.  $p \vee (p \wedge q)$
2.  $p \wedge (p \vee q)$
3.  $\neg(p \vee q) \vee p$
4.  $(p \wedge q) \vee (p \wedge \neg q)$
5.  $(p \rightarrow q) \wedge p$
6.  $(p \vee T) \wedge q$
7.  $\neg(\neg p \wedge q)$
8.  $(p \rightarrow q) \vee (p \rightarrow \neg q)$
9.  $(p \wedge F) \vee q$
10.  $(p \leftrightarrow q)$

Goal:

- Apply De Morgan
- Apply absorption
- Replace implication

### ◆ PART 3: 10 English $\rightarrow$ Predicate Logic Conversions

Let:

$S(x)$ : x is student

$P(x)$ : x passed

$E(x)$ : x is even

$N(x)$ : x is natural number

$L(x,y)$ : x loves y

Translate:

1. All students passed.
2. Some students failed.
3. Every natural number is positive.
4. There exists an even number.
5. Some students are intelligent.
6. No student failed.
7. Every student loves someone.
8. There exists a student who loves everyone.
9. Not all students passed.

10. There is no even number greater than 2 (in given domain).

Goal:

Use  $\forall$ ,  $\exists$  properly.

♦ **PART 4: 5 Negation of Quantified Statements**

Negate:

1.  $\forall x P(x)$
2.  $\exists x P(x)$
3.  $\forall x (P(x) \rightarrow Q(x))$
4.  $\exists x (P(x) \wedge Q(x))$
5.  $\forall x \exists y L(x,y)$

Use rules:

$$\neg \forall = \exists \neg$$

$$\neg \exists = \forall \neg$$